



Subject Code: 01CT0718

Subject Name: FPGA Based System Design

B.Tech. Year-IV (Semester VII)

Course Objectives:

The main objective of this course is to provide the basics about internal architecture of FPGA and working of FPGA.

Credits Earned: 04 Credits

Course Outcomes: After completion of this course, student will be able to:

1. Analyze system-level design methodologies including SoC, SoPC, IP-based design, and hardware-software co-design.
2. Apply top-down design approach and analyze data path, control path, and FSM (Mealy & Moore) designs for digital system development.
3. Analyze timing of sequential circuits, pipelining, and resource sharing techniques to optimize system performance and hardware utilization.
4. Analyze PLD and FPGA architectures.
5. Apply HDL-based design, FPGA programming, static timing analysis, and debugging techniques to develop efficient and reliable digital systems.

Pre-requisite of course: Basic Concept in digital circuit design

Teaching and Examination Scheme:

Teaching Scheme (Hours)			Credits	Theory Marks			Tutorial/ Practical Marks		Total Marks
				E	I		V	T	
Theory	Tutorial	Practical		ESE	IA	CSE	Viva	Term Work	
3	0	2	4	50	30	20	25	25	150

Included activities for assessment:

- MCQ test
- Assignments
- Lab Experiments
- Tutorials
- Application based simulation

Assessment scheme:

- Internal Assessment (IA): I and II Mid Terms (30 Marks each)

Total: 30 Marks (Average of I and II Mid Term Marks)

End Semester Exam (ESE)

- Total: 60 Marks
- Central level exam

(60 Marks will be converted into 50 Marks)

Continuous Semester Evaluation (CSE)

- 70 Marks: 7 Module wise quizzes.
- 50 Marks: 5 Assignments

(120 Marks will be converted into 20 marks)

Viva

- Total: 25 Marks
- 25 Marks: Oral exam from whole syllabus

Term work (TW)

- Total: 25 Marks
- 15 Marks: Lab Experiments & Lab Performance
- 10 Marks: Application based Simulation/Tutorials

Consolidate chart for assessment scheme:

Scheme	ESE	IA	CSE	Viva	Term Work (TW)
Marks	50	30	20	25	25
Evaluation Pattern	Central level exam	Central level exam	70 Marks: 7 Module wise quizzes. 50 Marks: 5 Assignments	25 Marks: Viva exam from whole syllabus	15 Marks: Lab Experiments & Lab Performance 10 Marks: Application based Simulation/Tutorials

Contents of Syllabus

Unit	Topics	Contact Hours
1	Revision of basic Digital systems Combinational Circuits, Sequential Circuits, Timing, Electrical Characteristics, Power Dissipation.	4
2	Current state of the field SoC, IP Design, SoPC, Design methodology, System Modeling, Hardware-Software Co-design, Device Technology, Application Domains.	6
3	Digital system Design. Top-down Approach to Design, Case study, Data Path, Control Path, Controller behavior and Design, Case study Mealy & Moore Machines. Timing of sequential circuits, Pipelining, Resource sharing.	6
4	Programmable Logic Devices. Introduction, Evolution: PROM, PLA, PAL, Architecture of PAL's, Applications, Programming PLD's, Design Flow, Programmable Interconnections, Complex PLD's (MAX - 7000, APEX). Architecture, Resources, Applications, Tools	8
5	FPGA Architecture Introduction, Logic Block Architecture, Routing Architecture, Programmable Interconnections, Design Flow, Xilinx Virtex-II Architecture, Altera Stratix Architecture	6
6	Programming the FPGA Boundary Scan, Programming of the FPGA, Constraint Editor, Static Timing Analysis, One hot encoding, Applications, Tools, Case Study, Xilinx Virtex II Pro, Embedded System on Programmable Chip. Hardware-software co-simulation, Debugging FPGA Design	6
7	HDL for Synthesis Introduction, Behavioral, Data flow, Structural Models, Simulation Cycles, Process, Concurrent Statements, Sequential Statements, Loops, Delay Models, Sequential Circuits, FSM Coding	6
Total Hours		42

Suggested Text books/Reference books:

1. Jon F Wakerly, Digital Design: Principles and Practices, Prentice Hall.
2. Kevin Skahil, VHDL for programmable logic, Addison Wesley.
3. Zainalabedin Navabi, VHDL, analysis and modeling of digital systems, McGraw-Hill.
4. Stephen Brown and Zvonko Vranesic, Fundamentals of Digital Logic with Verilog Design, McGraw Hill Third Edition
5. Seetharaman Ramachandran, "Digital VLSI Systems Design: A Design Manual for Implementation of Projects on FPGAs and ASICs Using Verilog", Springer Science & Business Media, 14-Jun-2007
6. PLD, FPGA data sheets

Suggested List of Experiments:

1. Introduction to the FPGA development board.
2. Interfacing input/output devices on the FPGA development board.
3. Implementing adder circuits such as half/full and ripple carry adder on the FPGA development board.
4. Implement multiplexers and demultiplexers on the FPGA development board
5. Implement encoders and decoders on the FPGA development board.
6. Implement code converters on the FPGA development board
7. Implement flip-flops and registers on the FPGA development board.
8. Implement a counter with a frequency divider on the FPGA development board.
9. Implement Mealy and Moore machine examples on the FPGA development board.
10. Implement a Traffic Light Controller on the FPGA development board.
11. Implement Digital Alarm Clock on the FPGA development board.
12. Implement 8-bit RISC processor on the FPGA development board.
13. Implement FIR/IIR filter on the FPGA development board.
14. Implement sensor-based systems on the FPGA development board.
15. Implement vending machine on the FPGA development board.

Capstone Mini-Project with Design Milestone

Structure (across the term)

Milestone	Activity	Weightage
M1	Problem Statement & Block Diagram	10%
M2	RTL Design & Simulation (Quartus/ModelSim/Vivado)	20%
M3	FPGA Implementation & Resource Report	30%
M4	Demonstration & Viva	30%
M5	Report / Documentation	10%

Suggested Mini-Project Topics (not in the experiment list)

Domain	Project Idea
Communication	UART / SPI / I ² C protocol controller
Image Processing	Edge detection / Histogram on VGA output
Audio	PWM-based audio synthesizer / Tone generator
Cryptography	AES encryption/decryption hardware accelerator
Embedded Control	PID controller for DC motor speed
Networking	Simple Ethernet packet sniffer (MAC layer)
Gaming	FPGA-based Pong / Snake game on VGA
Security	Fingerprint/RFID-based access control system
Robotics	Line-following robot controller
Medical	ECG signal acquisition and display



Major Equipment/software: FPGA board's IDE, Altera Quartus 13.0 Web Edition/ Xilinx Vivado

List of Open-Source Software/learning website:

NPTEL, Beginners Guide to get started with Xilinx FPGA Programming, FPGA4FUN.COM

Instructional Method:

1. Course delivery is the combined use of whiteboard/blackboard and PowerPoint presentation using a projector as per requirement.
2. Students will be assessed with a continuous evaluation approach with assignments and quizzes at regular intervals.
3. For effective communication with students, Google Classroom is used.

Supplementary Resources:

1. NPTEL- Digital System design with PLDs and FPGAs (<https://nptel.ac.in/courses/117108040>)
2. <https://nandland.com/learn-verilog/>
3. <https://chipverify.com/tutorials/verilog>
4. <https://www.fpga4fun.com/HDLtutorials.html>
5. <https://www.intel.com/content/www/us/en/docs/oneapi-fpga-add-on/developer-guide/2024-1/fpga-architecture-overview.html>